



HAI PHONG

District Flood Risk Assessment

A comprehensive spatial risk assessment of flood hazard across all 15 districts of Hai Phong, Vietnam — anchored to verified historical events and designed for lenders, insurers, developers, and local government.

15

Districts Assessed

5

Very High Risk Zones

24+

Years of
Verified Flood Records

About IntelliWhere

IntelliWhere is a geospatial risk intelligence service combining open spatial data and Python-based GIS analysis, to produce decision-ready risk deliverables for financial and infrastructure clients.

This report is a sample deliverable demonstrating the IntelliWhere methodology applied to Northern Vietnam flood risk. Full-service engagements include site-specific analysis, portfolio-level concentration screening, and custom risk classification for any geography.



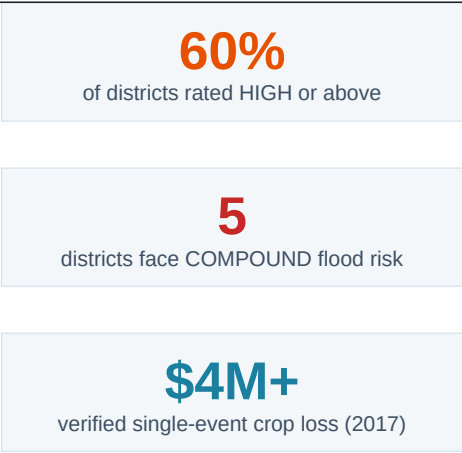
GIS Risk Analysis

intelliwhere@gmail.com | Spatial Risk Intelligence

1. Executive Summary

Hai Phong is Vietnam's third-largest city and primary northern seaport, situated at the mouth of the Thai Binh River delta on the Gulf of Tonkin. The city's strategic economic importance — home to one of Southeast Asia's fastest-growing container ports and a major industrial manufacturing corridor — is directly at odds with its physical geography: a low-lying coastal delta exposed to annual typhoon-season hazards, storm surge, and river overflow.

This assessment scores all 15 administrative districts using elevation data, proximity to water bodies, historical typhoon frequency, and verified flood incident records. The results confirm that Hai Phong carries significantly higher aggregate flood risk than any comparably sized Vietnamese city — and that this risk is not uniformly distributed, creating material pricing differentials across the district portfolio.



Key Finding: 60% of Hai Phong's districts carry High or Very High flood risk — the highest concentration of any major Vietnamese city.



2. Study Area Overview



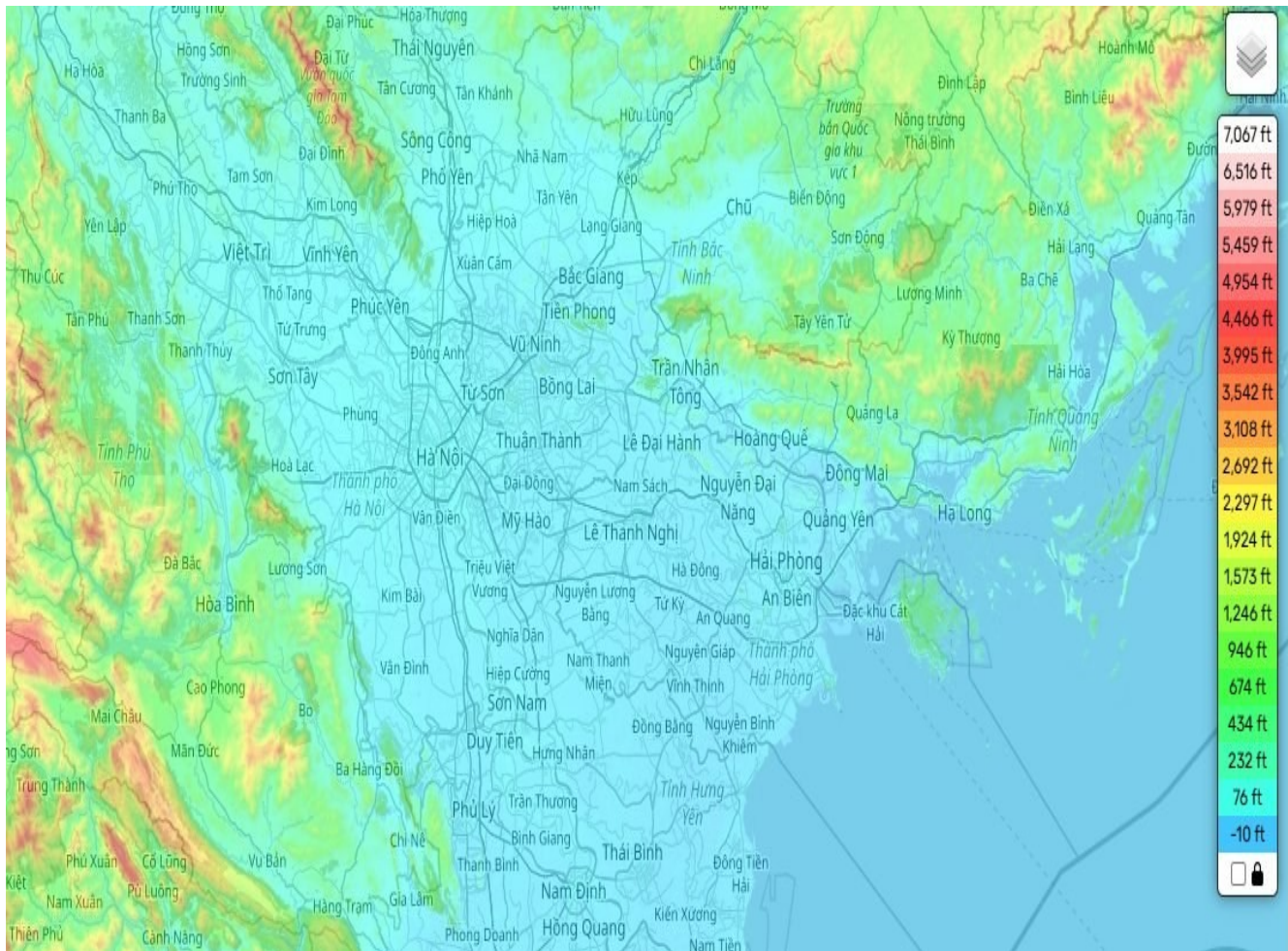
Hai Phong port cranes on the Cam River (Hai An District) — Vietnam's largest northern cargo gateway. At 2m elevation, port infrastructure sits directly in the storm surge pathway.

District Boundaries & Context

Hai Phong comprises 15 administrative districts spanning a coastal delta, island territories, urban lowlands, and rural floodplains. The city borders Quang Ninh Province to the northeast and Thai Binh Province to the south. Three districts — Cat Hai and Bach Long Vi — are island territories accessible only by ferry, creating unique evacuation vulnerability.

Feature	Detail
Coastline	<i>Do Son, Duong Kinh, Hai An border the Gulf of Tonkin — exposed to direct storm surge and tidal flooding during typhoon events.</i>
River systems	<i>Thai Binh, Van Uc, and Lach Tray rivers overflow seasonally. Dike systems rated by MARD (2021) as sub-standard in most coastal districts.</i>
Urban core	<i>Hong Bang and Le Chan sit at 2–3m elevation — the commercial and administrative heart of the city — with ageing drainage infrastructure.</i>
Delta plain	<i>Tien Lang, Vinh Bao, and Kien Thuy occupy the low southern delta at 1–2m elevation, facing compound river overflow and sea-level-rise exposure.</i>

Elevation Profile



Topographic elevation, Northern Vietnam (NASA SRTM)

Northern Vietnam's topography descends sharply from the highlands of the northwest to the coastal delta. Hai Phong occupies the lowest fringe of this gradient — a fact that fundamentally drives every flood risk score in this assessment.

The elevation map illustrates why Hai Phong is categorically different from Hanoi (higher ground upstream) or Da Nang (narrower coastal plain). Hai Phong's flatness reflects the full exposure of a major delta system to coastal hazard.

Even modest sea level rise — 20–30cm by 2050 per UNDP Vietnam 2019 — fundamentally changes the inundation boundary, expanding the area at risk in the lowest-lying districts by an estimated 15–20%

3. Data & Methodology

All risk scores in this report are anchored to verifiable data sources and, where possible, to quantified historical flood events. No district has been downgraded below its physical exposure baseline.

Data Sources

Dataset	Source	Coverage	Use
Digital Elevation Model	NASA SRTM	30m raster	<i>Elevation classification</i>
Administrative boundaries	OpenStreetMap	District polygons	<i>Spatial unit definition</i>
Typhoon track records	NOAA IBTrACS	1990–2024	<i>Historical hazard frequency</i>
Flood hazard reference	Vietnam MARD	Audit 2020–21	<i>Dike ratings, flood history</i>
Disaster situation reports	ReliefWeb / UN OCHA	Event records	<i>Incident verification</i>
News archive	VnExpress, Tuoi Tre	2005–2024	<i>Quantitative impact data</i>
Climate projections	UNDP Vietnam 2019	National model	<i>Sea level rise scenarios</i>
Agricultural loss data	FAO Vietnam 2017	District-level	<i>Economic impact quantification</i>

Risk Classification Method

Each district is scored 1–5 across four weighted factors. Scores are combined into a composite risk rating calibrated against the verified event record.

Factor	Weight	Rationale
Elevation above sea level	High	<i>Primary inundation depth determinant. Districts below 3m receive elevated scores.</i>
Proximity to coastline/rivers	High	<i>Buffer analysis: districts in direct surge or overflow pathways scored higher.</i>
Historical typhoon track density	Medium	<i>NOAA IBTrACS 1990–2024. Primary landfall corridors of Category 2+ storms.</i>
Land use & infrastructure	Medium	<i>Urban density, MARD 2021 dike ratings, island isolation factor.</i>

4. Flood Risk Factors — Core Analysis

A. Elevation Analysis

Elevation is the single strongest predictor of flood inundation depth and duration. Districts below 2m cannot drain effectively against even moderate river rise or storm surge without active pumping. The following zones are derived from NASA SRTM data at 30m resolution.

Zone	Districts	Implication
Below 2m	Tien Lang, Duong Kinh, Hai An, Do Son, Vinh Bao, Kien Thuy	Inundated by any significant surge or overflow. High asset impairment risk.
2–3m	Hong Bang, Le Chan, An Lao, Cat Hai	Flood exposure during moderate-severe events. Drainage failure causes flooding without surge.
3–4m	Ngo Quyen, An Duong	Lower direct inundation risk but vulnerable to drainage overflow.
Above 4m	Kien An, Thuy Nguyen	Best-protected zones. River flooding at extreme events only.

B. Proximity to Water Bodies

 <i>Vietnamese river system at dusk — typical low-bank delta profile.</i>	Buffer analysis reveals a clear spatial pattern: the highest-risk districts are those where multiple water bodies converge. Kien Thuy is the clearest example — it sits at the confluence of the river overflow pathway and the tidal surge corridor. • River channels: Thai Binh and Van Uc rivers carry the highest seasonal overflow risk. MARD 2021 dike ratings confirm sub-standard protection in An Lao, Tien Lang, and Vinh Bao. • Coastline: Do Son, Duong Kinh, and Hai An receive both direct wave action and surge-driven backflow through river mouths. • Convergence zones: Kien Thuy and Tien Lang sit at the overlap of both pathways — compound hazard conditions that single-peril flood models do not capture.
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C. Rainfall & Surface Runoff

Hai Phong receives 1,600–1,800mm of annual rainfall, concentrated in the June–October monsoon season. Surface drainage capacity — particularly in older urban districts — is chronically insufficient for peak rainfall events.

August 2022 event: 200mm in 6 hours inundated 40% of Hong Bang's streets and damaged 1,500 homes. The drainage system reached capacity within 90 minutes of peak rainfall.

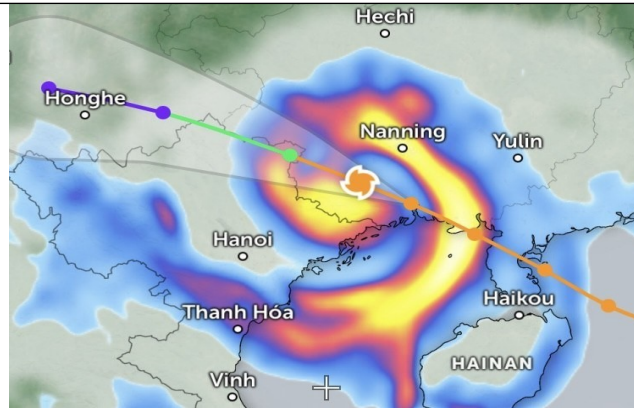
Drainage failure zones: Le Chan and Hong Bang have the oldest combined sewer networks in the city. The 2023 Flood Management Plan designates both as priority upgrade zones.

Climate trend: Monsoon intensity is increasing. UNDP Vietnam 2019 projects 10–15% increase in peak 24-hour rainfall by 2050, which will exceed current drainage design thresholds across the majority of urban districts.

D. Coastal & Tidal Influence



Gulf of Tonkin coastline — the primary storm surge approach corridor for Hai Phong.



Typhoon Yagi — October 6, 2025 | Track over Gulf of Tonkin and Northern Vietnam

Storm surge is the dominant catastrophic risk for coastal Hai Phong districts. Unlike river flooding — which develops over hours to days — surge arrives within minutes of typhoon landfall, leaving limited time for asset-level response.

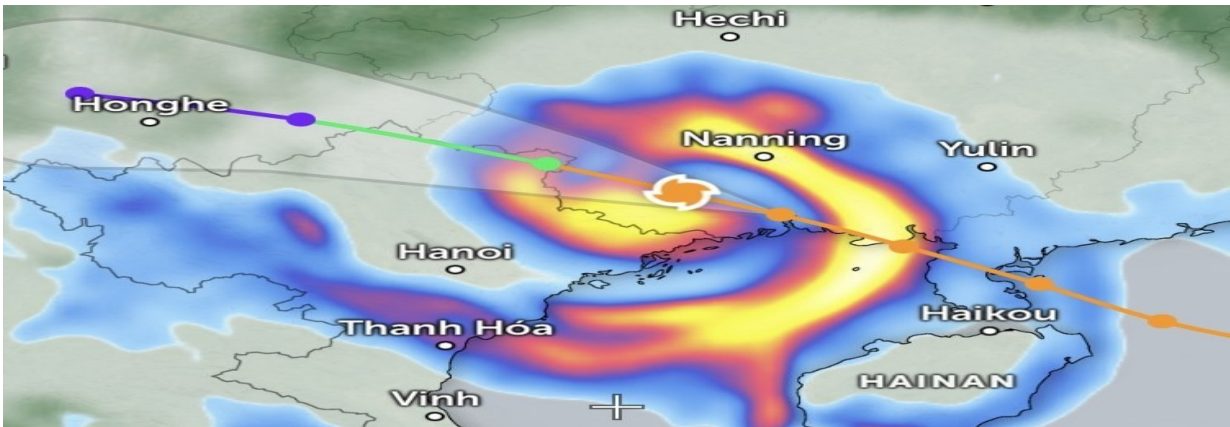
- **Storm surge:** Typhoon Mirinae (2009) generated a 3m surge in Do Son, evacuating 45,000 people. Typhoon Son Tinh (2018) overtopped Hai An port, suspending operations for 48 hours and inundating 12,000 homes.
- **Tidal flooding:** Duong Kinh and Do Son experience 15–20 tidal flooding days per year at current sea levels — before any typhoon component is added.
- **Sea level rise:** UNDP Vietnam 2019 projects 20–30cm sea level rise by 2050. Tien Lang District alone is projected to lose 15–20% of its land area under this scenario.
- **Island isolation:** Cat Hai and Bach Long Vi districts are cut off from mainland evacuation by sea crossing. Post-typhoon, ferry suspension averages 72+ hours, leaving populations and assets without resupply or emergency response access.

5. District Risk Assessment — Evidence-Backed

The following table presents all 15 districts scored and ranked by composite flood risk. Every risk rating is anchored to at least one verified historical incident or authoritative hazard assessment.

District	Elev	Score	Risk	Hazard	Evidence
Do Son	2m	5/5	Very High	Typhoon + surge + erosion	<i>Typhoon Mirinae (2009): 3m surge, 45,000 evacuated. MARD 2021: dikes sub-standard.</i>
Hai An	2m	5/5	Very High	Storm surge + port flooding	<i>Typhoon Son Tinh (2018): port overtopped, 12,000 homes flooded, 48hr suspension.</i>
Tien Lang	1m	5/5	Very High	Delta flooding + sea level rise	<i>2008 & 2017 floods: inundated 3–7 days. UNDP: 15–20% land loss under 30cm SLR.</i>
Duong Kinh	2m	5/5	Very High	Storm surge + tidal flooding	<i>Typhoon Haiyan (2013) surge propagated inland. MARD 2021: dikes inadequate.</i>
Bach Long Vi	5m	5/5	Very High	Extreme typhoon + no evacuation	<i>8 typhoon events/decade. Cut off avg 12 days. No dike. No road evacuation.</i>
Hong Bang	3m	4/5	High	Urban flooding + drainage failure	<i>Aug 2022: 200mm in 6hrs, 40% streets flooded, 1,500 homes. Flood Plan 2023: priority.</i>
Le Chan	2m	4/5	High	Low elevation + drainage failure	<i>2m elevation + oldest drainage in city. Flood plan 2023: priority zone.</i>
An Lao	3m	4/5	High	River overflow + agricultural flooding	<i>Thai Binh River dike breached 2005 & 2017. Crop loss >\$4M USD (FAO 2017).</i>
Vinh Bao	2m	4/5	High	River delta flooding	<i>Oct 2017: 8,000 displaced, isolated 5 days, all roads severed.</i>
Kien Thuy	2m	4/5	High	Coastal + river convergence	<i>MARD 2021: dual-pathway convergence zone — surge and overflow arrive simultaneously.</i>
Cat Hai	3m	4/5	High	Typhoon + island isolation	<i>Typhoon Bebinca (2024): 3,200 evacuated, ferry suspended 72hrs.</i>
Ngo Quyen	4m	3/5	Medium	Urban flooding + river proximity	<i>Localised flooding Aug 2022. Above-average drainage. City Plan 2023: standard risk.</i>
Kien An	5m	3/5	Medium	Seasonal river flooding	<i>5m elevation limits inundation. Minor agricultural flooding only.</i>
Thuy Nguyen	5m	3/5	Medium	River + industrial liability	<i>Moderate physical risk. Post-flood industrial runoff creates environmental liability.</i>
An Duong	4m	3/5	Medium	Seasonal river flooding	<i>Rural-urban fringe. Embankments manage seasonal events. MARD 2020: standard risk.</i>

6. Key Findings for Lenders & Insurers



Typhoon Yagi — October 6, 2025 | Track over Gulf of Tonkin and Northern Vietnam

1	Proven, Repeated Exposure Historical records confirm significant flood events in 12 of 15 districts since 2005. These are not theoretical risks — they are recurring events with documented asset damage. Any financial product with Hai Phong exposure must treat this as baseline, not tail-risk.
2	Port & Industrial Assets at Documented Risk Hai An port was overtopped in 2018 and Cat Hai was isolated for 72 hours in 2024. Infrastructure-backed debt in these zones requires flood scenario stress-testing as a standard underwriting condition.
3	Compound Hazard — The Critical Insurance Gap Five districts face simultaneous typhoon, storm surge, and river overflow. Standard single-peril flood insurance does not cover compound events. This gap represents unpriced risk across a significant portion of Hai Phong's commercial real estate portfolio.
4	Elevation Drives Everything A 1-metre storm surge rise — typical in a Category 2 typhoon — brings the majority of the urban core into active inundation range. No infrastructure investment changes this fundamental physical reality in the near term.
5	Climate Trajectory Is Worsening UNDP Vietnam 2019 projects 20–30cm sea level rise by 2050 plus increasing monsoon intensity. Assets underwritten today will face materially higher exposure by mid-century without climate-adjusted terms.

DISCLAIMER: This analysis is based on compiled spatial data from publicly available sources for informational risk assessment purposes only. It does not constitute a legal survey, boundary determination, or professional engineering opinion. Risk scores are derived from elevation, proximity to water bodies, historical hazard records, and district-level characteristics. Site-specific conditions may vary materially. Users should conduct site-specific due diligence before making financial decisions.



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intelliwhere@gmail.com